

DR#1 — Enterprise Knowledge Assistant

What I'm building. I am building TechNova Solutions' Enterprise Knowledge Assistant (EKA), a narrowly scoped employee question-and-answer service over approved internal organisational knowledge. The assistant will retrieve existing policy and process content, answer with source references and a confidence indicator, and explicitly escalate when the corpus does not contain sufficient evidence. It will not answer public-web questions, create policies, make approvals, execute transactions, or replace HR, IT, Security, Procurement, Privacy, or managers.

For whom. Priya Nair, an HR Generalist, needs fast policy answers with citations while supporting employees. Arjun Mehta, a Software Engineer, needs self-service for IT, access, security, and support procedures. Kavita Rao, a Procurement Specialist, needs traceable approval and supplier guidance. Daniel Thomas, an Operations Manager, needs consistent answers before advising his team. Meera Shah, a Compliance/Privacy Partner, needs cautious answers and reliable escalation for higher-risk questions.

Architecture today. The M1 architecture is deliberately a thin FastAPI foundation. A Bearer-token protected /ask endpoint accepts a question and returns the locked six-field Answer schema: content, cost_usd, retries, confidence, sources, and schema_version. Logging and cost-tracking middleware are already separated from the handler, and unit tests exercise authentication, validation, and response shape. The current handler returns a placeholder because M1 is about framing and evaluation readiness; real retrieval is planned for the next phase. The target architecture is Employee → API → Retriever → grounded LLM → cited Answer.

KPI targets. Question resolution rate has a target of at least 80% of answerable golden questions; this establishes useful self-service while preserving escalation for unsupported cases. Retrieval/source accuracy targets at least 90% correct source selection on answerable questions because an EKA must retrieve the right organisational evidence, not merely produce plausible text. Answer latency targets at least 90% of answered requests within 30 seconds because the business objective is to reduce waiting for HR, IT, and subject-matter support. Operational numeric baselines will be instrumented during the pilot because M1 does not contain measured production data.

Trade-offs already made. First, I chose a controlled internal corpus rather than a general LLM because governance and citation are more important than open-domain coverage. Second, I chose one stable REST /ask endpoint rather than separate cheap/quality endpoints so model and retrieval choices remain internal. Third, I accept deliberate refusal for unsupported questions because confidently inventing policy is a more serious failure than escalating.

Uncertainties and risks. Retrieval quality, chunking, and confidence calibration are not yet validated. The corpus is synthetic for the capstone and must be replaced or formally approved before production use. Identity, authorization, audit retention, document versioning, and measured KPI baselines require later design.